

A change-time approach to temporal adverbs

Temporal adverbs like *already* are argued to presuppose that their propositions hold or do not over time intervals before or after the topic/reference time (Löbner 1989, Krifka 2000, Ippolito 2007). This paper proposes a contrary argument based on fieldwork and narratives from the Kiowa language, that these adverbs are tense adverbs. They presuppose a temporal relation between the topic time and a change time, the time of an anaphoric change situation salient in the context. In this sense, they work like perspectival *now/then*, but without a deictic component. **Kiowa /hègǎ/**. The Kiowa (kio) adverb /hègǎ/ corresponds to a broad set of translations, as the sequences in (1-4) demonstrate (glosses simplified for exposition).

(1) 'already'	/á:dò dè=tʰón/ stick I.it=found	/hègǎ è=têm+dò:/ already it=break+be	'I found a stick. It was already broken.'
(2) 'then'	/á:dò dè=tʰón/ stick I.it=find.PFV	/hègǎ dè=têm/ then I.it=broke	'I found a stick. Then I broke it.'
(3) 'now'	/á:dò dè=têm/ stick I.it=break.PFV	/hègǎ è=têm+dò:/ now it=break+be	'I broke a stick. Now it is broken.'
(4) 'no more'	/á:dò nó=dó:, né/ stick it.to me=be but	/hègǎ hón nó=dó:-mô:/ no more not it.to me=be-NEG	'The stick was mine, but it isn't anymore .'

Sometimes, /hègǎ/ appears multiple times in a sentence, as in (5), where the narrator is talking about how late he was working one night, setting the stage for a spectral visitor.

(5) /pàj+êl clock+big I.it=looked at	gǎ=bó: and	nò hègǎ now	gǎ:kʰòp'ǵj+tʰòp night+mid+past	hègǎ already	Ø=tʰó:gǎj it=pass.PFV	and so	nò hègǎ I.them=stop+prepared	gǎt=pʰátkǎ+mò:hòl/ 'I looked at the big clock and it was already past midnight so I prepared to stop.'
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Generalization. To form a generalization, we will not start by applying analyses in the literature, because 'already' and 'no more' involve presuppositions (e.g. 'no more' presupposes that p held in the past), while these uses of 'now' and 'then' are deictic to a narrative time. Instead, we try to capture what these all have in common in Kiowa that leads to the same morpheme being used.

The common link is that in each case, the topic time comes in the aftermath of some other relevant change. For instance, in (1) you found a stick (at t_1), and it became broken before you found it, at t_0 . In fact, the form of 'break' with 'be' (/têm+dò:/) in (1) denotes a result stative.

clause 1	change situation	clause 2
(1') t_1 I found a stick	s_0 it broke	t_1 it was already broken
(2') t_1 I found a stick	s_1 I found it	t_2 then I broke it
(3') t_1 I broke a stick	s_1 I broke it	t_2 now it is broken
(4') t_1 The stick was mine	s_2 I lost it	t_3 it isn't mine anymore

In each case the topic time in the clause with /hègǎ/ follows a change-time, the time of the change situation. The difference in English translation reflects that situation. If it is the previously mentioned situation, we get narrative progression to 'now' or 'then'. If the change effects a flip from a property holding to it not holding, 'no more' results. This change can be anaphoric to the preceding clause, or some other time. If the anaphor is contextually relevant, and the proposition holds, 'already' results.

In addition, /hègǎ/ is very commonly used in sentences translated with a perfect; Kiowa does not have perfect morphology. In this case, the anaphoric situation can be the start of the perfect interval, in which the topic time lands.

(6) /jǎ:-dò two-times France-to	France-kù hègǎ à=bá:/ have I=went	(7) /hègǎ háòtè+kʰì: have several+day I.it=be coughing	gǎ=kʰînmò/ 'I've been coughing for several days.'
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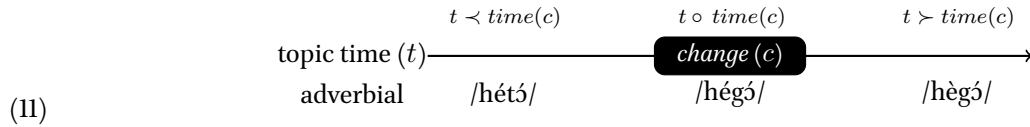
Proposal. The proposed analysis builds off this generalization. /hègǎ/ expresses a temporal succedence relation between some situation pronoun (c) and another time (i) (8a). This relation is presupposed, since it projects out of common triggers like negation (4) and conditional clauses. Applied to a change situation (like s_0) in (8b), the result is a property of times for the topic time to saturate.

- (8) a. $\llbracket \text{hègǎ} \rrbracket^g = \lambda c \lambda i. g(c)$, only if $i \succ \text{time}(g(c))$: b. $\llbracket \text{hègǎ} \rrbracket^g(s_0) = \lambda i. g(0)[i \succ \text{time}(g(0))]: \langle i, t \rangle$

Other relations. The succedence relation raises the question of other relations, and these are expressed by other

temporal adverbs. /hétó/ ‘still’ expresses that the topic time precedes an anaphoric change situation ($time(s_2)$ in (9)), which may not ever happen. With negation, ‘still not’ is used rather than ‘not yet’. The adverb /hégó/ expresses that the topic time overlaps or immediately follows the change (10); Watkins (1984) and McKenzie et al. (2022) gloss it as ‘just as’. The relations are schematized in (11).

- (9) é:hò:=àl t_1 **hétó** hón bét=t’á:già+òmgo:-hèl $t_1 \prec time(s_2)$
 now=even t_1 **still** not by.them=right+be made.NEG-EVID
 ‘even now they have not gotten things straightened out yet.’
- (10) /hégó/ t_1 hón hájá é=ó:mô:/ /hégó/ t_2 é=t’ó:tó: gò pòj k’i:già tsô: bàt=ôj+òmto:/
 then t_1 not at all he.to me=did.NEG **just as** he.to me=spoke & not later thus you.it=again+will do
 ‘Then he didn’t do anything to me. And he was telling me “Don’t do it again.”’ (McK. et al 2022)(S149)



Under this approach, these tense adverbials resemble *now* and *then* in narratives (Stojnic & Altshuler 2021). Anand & Toosarvandani (2019) argue that these are temporal demonstratives, extending their deixis to an assessment (a) rather than the utterance or topic time. The deictic time (d) is either in the time of a (*now*) or out of it (*then*).

- (12) *She was now a senator.* ($d \subseteq time(a)$) (13) *She then became a senator.* ($d \not\subseteq time(a)$)

However, the Kiowa tense adverbials here cannot be deictic to the utterance. To describe ‘at this/that time’ deictic adverbs are used: /é:hò:/ ‘now’ (9), /ójhò:/ ‘then’, and /ôjhò/ ‘right now/then’. We can conclude that elements of narrative cohesion (like *now*) can converge functionally from distinct formal mechanisms.

Extension. The change-time approach helps understand English *already*. Löbner (1989) claims that *already p at t* presupposes that p was false before t . Krifka (2000) and Ippolito (2007) claim instead that it presupposes that p is true at some point after t (14). However, that presupposition can be countered, too (15-16).

- (14) A: *Is your husband applying for citizenship?* B: *He is already American, since he was born there.*
 (15) *When I wrested the gun from the assailant’s hands, he was already aiming it at me.*
 (16) *She was already a senator when the scandal forced her resignation from public life.*

The change-time account avoids these pitfalls— in each case, some relevant change-time precedes the topic time, and the time p holds at is not relevant to *already*. In (14) the change is gaining citizenship; in (15) the start of aiming, in (16) becoming a senator. *Already* also often connotes a sense of violated expectation, and that can be inferred pragmatically if some milestone change precedes the topic time (17), but as (16) shows, the inference is not required.

- (17) *She was only 32 but was already a senator.*

This approach can likely be ported to non-temporal scales, to derive marginality readings (18), where the change to a property is lower on the scale than the degree in mind.

- (18) *Flying is already dangerous, and you wanna go skydiving?*

Conclusion. Ultimately, the change-time approach works very well to derive the breadth of the Kiowa adverb /hégó/, which is the most common word in the language. This approach tells us that temporal adverbs need not be deictic, and do not have to presuppose anything about when the proposition holds. It also shows that tense-like relations can be used outside of the grammatical tense system.

One added benefit emerges with fieldwork. Doing elicitation with a language as endangered as Kiowa, one sometimes encounters attrition. Eliciting these kinds of words can sometimes inadvertently make speakers worry about attrition, since there is no single word for ‘already’, or ‘anymore’. This paper’s result helps demonstrate that the meanings do get expressed, and that languages may use a variety of formal mechanisms to express them.

References. Anand & Toosarvandani (2019). Now and then: Perspectives on positional variance in temporal demonstratives. *SuB* | Krifka (2000). Alternatives for aspectual particles: Semantics of *still* and *already*. *BLS* | Löbner (1989). German *schön*, *erst*, *noch* An integrated analysis. *L&P* | Ippolito (2007). On the meaning of some focus-sensitive particles. *NALS* | McKenzie et al. (2022). *Plains Life in Kiowa ...* | McKenzie, P. (n.d). *Ghost Woman* (ms.) | Stojnic & Altshuler (2021). Formal properties of *now* revisited. *S&P* | Watkins (1984). *A Grammar of Kiowa*